UNIT - V: DEEP LEARNING

Autoencoders & VAE, Transformers, GPT, Generative and Agentic AI | 2 Marks

Q1. Define Variational Autoencoder (VAE). What are its primary structural components?

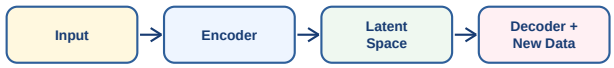
Ans:

A **Variational Autoencoder (VAE)** is a generative model that learns the data distribution and generates new data similar to the training data.

It has three main components:
1. **Encoder:** Converts input data into a smaller representation.
2. **Latent Space:** Stores the compressed representation of data.
3. **Decoder:** Generates new data from the latent representation.

Easy memory: Encoder → Latent Space → Decoder.

Simple sketch:



Q2. What are Query (Q), Key (K), and Value (V) in the Transformer Self-Attention mechanism?

Ans:

Query, Key, and Value are the three main elements of the Self-Attention mechanism in Transformers.

1. **Query (Q):** Represents what information a token is looking for.
2. **Key (K):** Helps to find which other tokens are important or relevant.
3. **Value (V):** Contains the information that is combined using attention weights.

Easy memory: Q = What to find. K = Which is relevant. V = Information.

Simple sketch:

Q3. What is an AI Agent? List the fundamental building blocks of an Agentic AI system.

Ans:

An **AI Agent** is a system that observes its environment, makes decisions, and performs actions to achieve a goal.

Main building blocks are:
1. **Perception:** Receives information from the environment.
2. **Reasoning and Planning:** Decides what to do.
3. **Memory:** Stores useful information.
4. **Tools and Actions:** Performs tasks.
5. **Feedback and Learning:** Improves its performance using results.

Simple sketch:

Q4. What is a Generative Adversarial Network (GAN)? Name its two competing sub-networks.

Ans:

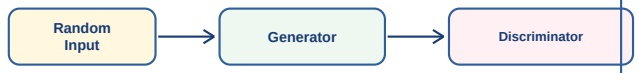
A **Generative Adversarial Network (GAN)** is a generative model that contains two neural networks which compete during training.

Its two sub-networks are:
1. **Generator:** Creates fake data similar to real data.
2. **Discriminator:** Identifies whether the given data is real or fake.

Both networks compete to improve the quality of generated data.

Easy memory: Generator = Creates fake data. Discriminator = Checks real or fake.

Simple sketch:



Q5. Mention any four real-world applications of Generative AI across different industries.

Ans:

Four applications of Generative AI are:

1. **Text Generation:** Creates essays, stories and summaries.
2. **Image Generation:** Creates images from text descriptions.
3. **Audio Generation:** Generates music and human-like speech.
4. **Code Generation:** Creates computer programs and helps developers write code.

Q6. What does GPT stand for, and what fundamental neural architecture powers it?

Ans:

GPT stands for **Generative Pre-trained Transformer**.

1. It is a type of AI language model.
2. It is based on the Transformer architecture.
3. It generates text by predicting the next token based on previous tokens.

Easy memory: GPT = Generative + Pre-trained + Transformer.

Simple sketch:

UNIT - V: CONTINUED

Q7. Name any two prominent NLP applications of Transformer models.

Ans:

Two important NLP applications of Transformer models are:

1. **Machine Translation:** Converts text from one language to another.
2. **Text Summarization:** Converts long text into a short and meaningful summary.

Other applications include question answering and text generation.

Q8. What is Generative AI? How does it differ fundamentally from Discriminative AI?

Ans:

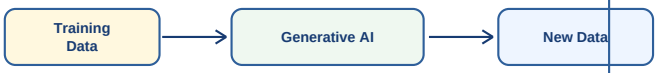
Generative AI is a type of AI that learns patterns from existing data and creates new data such as text, images and audio.

Difference:

Generative AI	Discriminative AI
Creates new data.	Classifies existing data.
Learns data patterns.	Learns differences between classes.
Example: ChatGPT, image generation.	Example: Spam detection.

Easy memory: Generative = Creates. Discriminative = Classifies.

Simple sketch:



Q9. What is Next-Token Prediction in autoregressive language models?

Ans:

Next-Token Prediction is a process in which a language model predicts the **next word or token** based on the previous tokens.

1. The model receives input text.
2. It predicts the next token.
3. The predicted token is added to the text.
4. This process repeats to generate a complete sentence.

Example: Input: "I love" → Predicted next token: "India" → Output: "I love India."

Simple sketch:

Q10. What is the reparameterization trick in Variational Autoencoders and why is it needed?

Ans:

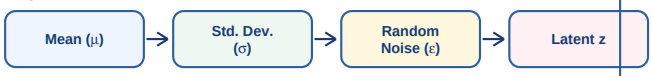
The **Reparameterization Trick** is a method used in VAE to generate a latent vector using the mean and variance.

Formula: $z = \mu + \sigma \times \varepsilon$

Where:
- **z** = Latent vector
- μ = Mean
- σ = Standard deviation
- ε = Random noise

Why is it needed?
1. It allows the VAE to generate random samples.
2. It helps gradients flow during backpropagation.
3. It helps the encoder and decoder learn properly.

Simple sketch:



QUICK REVISION

Q.No	Important Point
Q1	VAE = Encoder + Latent Space + Decoder
Q2	Q = Looking, K = Relevance, V = Information
Q3	AI Agent = Perception, Planning, Memory, Tools, Feedback
Q4	GAN = Generator + Discriminator
Q5	Text, Image, Audio, Code Generation
Q6	GPT = Generative Pre-trained Transformer
Q7	Machine Translation + Text Summarization
Q8	Generative = Creates, Discriminative = Classifies
Q9	Predicts the Next Token
Q10	Reparameterization = Mean + Variance + Random Noise